LangChain – Assignments

Python + LangChain + OpenAI

Technical requirements:

- Python
- LangChain
- LangChain OpenAI
- OpenAI GPT model (GPT-5.5)
- Python dotenv
- Reusable prompt templates
- No chains required

Assignment 1: Smart Email Generator

Scenario: Build an application that generates professional emails based on user input.

Requirements:

1. Accept recipient role, email purpose, and key points from the user.
2. Create a reusable prompt template.
3. Generate a professional email using the LLM.
4. Allow the user to select the email tone:
    - Formal
    - Friendly
    - Persuasive
5. Display the generated email.

Expected output:

```
Recipient: Project Manager
Purpose: Request project deadline extension
Tone: Formal

Subject: Request for Project Deadline Extension

Dear Project Manager,

I would like to request an extension...
```

Assignment 2: Python Code Reviewer

Scenario: Develop an AI-based Python code review assistant.

Requirements:

1. Accept Python code from the user.
2. Create a reusable prompt template.
3. Ask the LLM to review the code.
4. Identify:
    - Coding mistakes
    - Logical errors
    - Performance issues
    - PEP 8 violations
5. Suggest an optimized version of the code.
6. Explain the improvements.

Expected output:

```
Code Review Report

1. Issues Identified:
   - Unnecessary loop
   - Poor variable naming

2. Optimized Code:
```

```
    [Improved Python code]

3. Explanation:
   [List of improvements]
```

Assignment 3: AI Interview Preparation Assistant

Scenario: Create an AI application that generates technical interview questions.

Requirements:

1. Accept technology name (Python, SQL, Java, etc.).
2. Accept experience level (Beginner, Intermediate, Advanced).
3. Accept the number of questions.
4. Create a reusable prompt template.
5. Generate interview questions.
6. Generate answers and explanations.

Additional challenge: Generate practical coding questions along with theoretical questions.

Assignment 4: Product Description Generator

Scenario: Build an AI-based product description generator for an eCommerce application.

Requirements:

1. Accept product name.
2. Accept product category.
3. Accept product features.
4. Accept target audience.
5. Accept description style.
6. Generate an attractive product description using a prompt template.

Additional challenge: Generate three different versions:

- Short description
- Detailed description
- SEO-friendly description

Assignment 5: AI SQL Query Assistant

Scenario: Develop an application that converts natural-language requirements into SQL queries.

Requirements:

1. Accept database type (SQL Server, MySQL, SQLite).
2. Accept table structure.
3. Accept user requirements in plain English.
4. Create a reusable prompt template.
5. Generate the SQL query.
6. Explain the generated query.

Example input:

```
Database: SQL Server

Table: Employees
Columns:
Id, Name, Department, Salary

Requirement:
Display employees whose salary is
greater than 50000.
```

Expected output:

```
SELECT *
FROM Employees
WHERE Salary > 50000;
```

Assignment 6: AI Customer Support Assistant

Scenario: Build an AI-powered customer support assistant for an eCommerce application.

Requirements:

1. Accept customer queries.
2. Accept the company/product information as context.
3. Create a reusable prompt template.
4. Generate responses based on the supplied context.
5. Maintain a polite and professional tone.
6. If information is unavailable, the assistant should clearly state that it doesn't know rather than inventing an answer.

Additional challenge: Allow users to select response styles: Short, Detailed, and Friendly.

Assignment 7: Resume Analyzer

Scenario: Develop an AI application that analyzes resumes against a job description.

Requirements:

1. Accept resume content as text.
2. Accept job description.
3. Create a reusable prompt template.
4. Analyze the candidate's skills.
5. Identify matching and missing skills.
6. Provide improvement suggestions.
7. Generate an overall suitability score with justification.

Expected output:

```
Resume Analysis

Matching Skills:
- Python
- SQL
- REST API

Missing Skills:
- Docker
- AWS

Suitability Score: 75/100

Recommendations:
- Improve cloud knowledge.
- Add practical project experience.
```

Assignment 8: AI Learning Assistant

Scenario: Develop an AI-powered learning assistant that explains technical concepts.

Requirements:

1. Accept a technical topic.
2. Accept the learner's knowledge level.
3. Accept explanation style.
4. Create a reusable prompt template.
5. Generate:
    - Concept explanation
    - Real-world example
    - Python code example (where applicable)
    - Three practice questions

Additional challenge: Allow the user to request a simplified explanation of the same topic.

Common technical requirements for all assignments

application structure:

```
langchain_assignments/
│
├── main.py
├── prompt_templates.py
├── model_config.py
├── .env
└── requirements.txt
```

Suggested reusable prompt template structure:

```
from langchain_core.prompts import PromptTemplate

prompt = PromptTemplate(
    input_variables=["topic", "level"],
    template="""
    Explain {topic} for a {level} level learner.
    Include examples and practical applications.
    """
)
```